

PAIR-Grasp: A Novel Benchmark Dataset and Model for Low-Light Robotic Grasp Detection

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Abstract—Reliable robotic grasping under low-light conditions remains challenging because weak signals, sensor noise, depth degradation, and specular reflections obscure object boundaries and grasp-relevant geometry. Existing solutions often rely on image enhancement or additional sensing modalities, introducing artifacts or system complexity. Inspired by human perception under poor illumination, we propose PAIR-Grasp, a physics-grounded framework using only standard RGB-D observations. It extracts illumination-invariant structural cues from degraded RGB images and adaptively incorporates them into multi-scale RGB-D representations, suppressing illumination interference while preserving grasp-relevant geometry. We also construct RLL-Grasp, the first real-world low-light RGB-D grasp detection dataset, covering diverse viewpoints, illumination levels, and backgrounds. PAIR-Grasp achieves 96.0% detection accuracy on RLL-Grasp and a 91.7% success rate in real-world robotic grasping trials, demonstrating its effectiveness in challenging low-light environments.

I. INTRODUCTION

Robotic grasping enables embodied agents to physically interact with the environment and relies on visual observations to locate graspable regions and estimate executable grasp configurations [1]. Although existing methods achieve high accuracy under normal illumination, their reliability deteriorates under low illumination and shadowed conditions [2], with sensor noise, depth degradation, and specular reflections further corrupting visual observations. As shown in Fig. 1, these degradations obscure object boundaries and contact-relevant geometry, leading to inaccurate grasp localization and orientation estimation [3], [4]. Therefore, low-light robotic grasping is not merely an image-enhancement problem, but a perception-to-action challenge.

Existing solutions often employ image restoration [5], [6] or additional sensors [2]. The former may introduce artifacts, whereas the latter increases system complexity and requires precise spatiotemporal alignment. Inspired by the human ability to perceive and manipulate objects under poor illumination, we propose PAIR-Grasp, a physics-grounded framework that extracts illumination-invariant cues from degraded RGB-D observations to improve grasp robustness under challenging lighting conditions. Furthermore, we construct RLL-Grasp, the first real-world low-light RGB-D grasp detection dataset, to evaluate robotic grasping under diverse illumination degradations and reflective interference.

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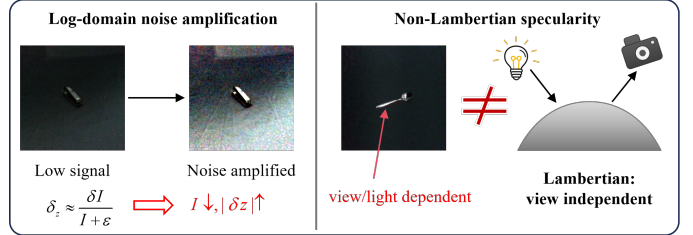


Fig. 1. Key challenges in low-light RGB-D grasp detection. In dark regions, log mapping amplifies low-signal perturbations, while view- and illumination-dependent specular highlights violate the Lambertian assumption underlying chromatic invariance.

II. METHODOLOGY

PAIR-Grasp comprises Physics-Grounded Illumination Decoupling (PID) and Attention-Gated Feature Modulation (AGFM). PID extracts illumination-invariant cues from degraded RGB via structure-residual decomposition and diffuse-like/specular-like separation, while AGFM injects them into multi-scale RGB-D representations through gated affine modulation, preserving grasping-relevant geometry.

A. Physics-Grounded Illumination Decoupling

In low-signal regions, logarithmic mapping amplifies observation noise. For the stabilized log response $z_c(p) = \log(I_c(p) + \epsilon)$, this effect is characterized by

$$\Delta z_c(p) = \log \left(1 + \frac{\eta_c(p)}{I_c(p) + \epsilon} \right), \left| \frac{\partial z_c(p)}{\partial I_c(p)} \right| = \frac{1}{I_c(p) + \epsilon}, \quad (1)$$

where $I_c(p)$ and $\eta_c(p)$ denote observed intensity and noise, respectively. PID therefore performs structure-residual decomposition before logarithmic mapping and separates diffuse-like and specular-like latent components to stabilize invariant extraction.

From the diffuse-like branch, PID applies multi-scale reflected convolutions to pairwise log-chromatic differences:

$$O_{i,j}^{(k)} = Z_i * K^{(k)} + Z_j * (-K^{(k)}) = (Z_i - Z_j) * K^{(k)}, \quad (2)$$

where Z_i and Z_j are stabilized log responses, and $K^{(k)}$ is a 3×3 or 5×5 zero-mean filter bank. The resulting responses are fused into the illumination-invariant feature F_{iiim} .

B. Attention-Gated Feature Modulation

AGFM uses backbone-guided spatial attention to gate the scale-aligned F_{iiim} and predict scale-specific affine parameters:

$$\hat{C}_i = C_i + \gamma_i \odot C_i + \beta_i, \quad (3)$$

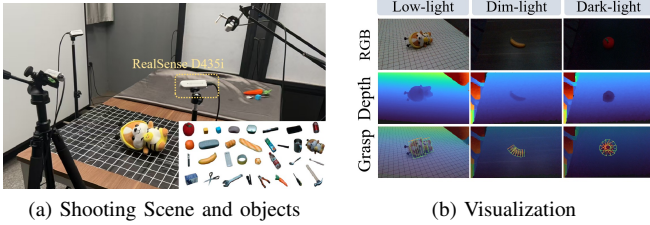


Fig. 2. Construction of the real-world low-light RGB-D grasp dataset RLL-Grasp. (a) RGB-D acquisition setup and object set. (b) Representative RGB, depth, and grasp annotations under different illumination levels.

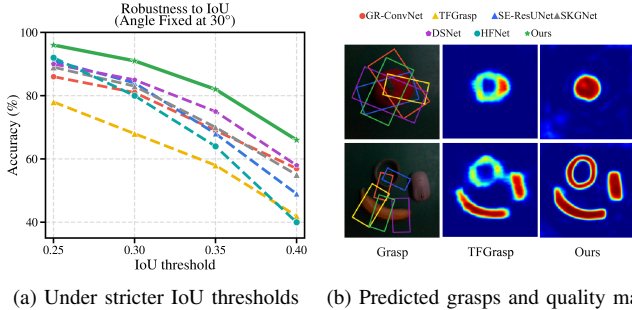


Fig. 3. Robustness and qualitative evaluation under real low-light conditions. (a) Grasp detection accuracy on RLL-Grasp under increasingly strict IoU thresholds with the angle threshold fixed at 30°. (b) Comparison of predicted grasps and quality maps in single- and multi-object scenes.

where C_i and $\hat{C}_i$ are the original and modulated RGB-D features, respectively. The modulated features are decoded into grasp quality, angle, and gripper-width maps.

III. PRELIMINARY EVALUATION

A. RLL-Grasp Dataset

We construct RLL-Grasp, a novel real-world low-light RGB-D grasp detection dataset with 3,000 samples covering 32 household and workshop objects across four viewpoints, four backgrounds, and three illumination levels of approximately 51, 17, and 7 lux. Fig. 2 shows the acquisition setup and representative samples.

Unlike most grasp detection benchmarks (e.g., GraspNet-1Billion) that are collected under well-lit conditions, and even compared to existing synthetic low-light grasp datasets, RLL-Grasp records more diverse, real-world degradations including specular reflection and high sensor noise. These features make RLL-Grasp a more challenging benchmark for evaluating the cross-illumination generalization and anti-interference capability of existing grasp detection algorithms.

B. Quantitative Results

Using the standard rectangle metric, PAIR-Grasp achieves 96.0% grasp detection accuracy on RLL-Grasp, outperforming SE-ResUNet (92.3%) by 3.7 percentage points. We further evaluate robustness by progressively increasing the IoU threshold while fixing the angle threshold at 30°. As shown in Fig. 3 (a), PAIR-Grasp maintains relatively high detection accuracy under stricter grasp localization criteria.

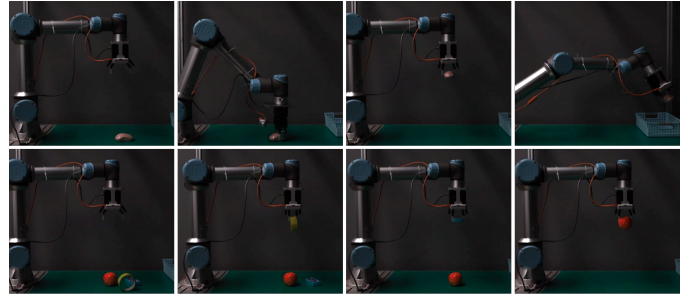


Fig. 4. Real-world low-light robotic grasp execution sequences in single- and multi-object scenes.

C. Qualitative Analysis

Fig. 3 (b) shows qualitative results in real-world low-light single- and multi-object scenes. PAIR-Grasp produces compact and well-localized grasp-quality responses, and the predicted grasps and quality maps remain aligned with geometrically favorable grasp regions.

D. Real-World Grasping Experiments

We further evaluate PAIR-Grasp in real-world robotic grasping experiments under low-light conditions. Fig. 4 presents representative grasp execution sequences in both single- and multi-object scenes. PAIR-Grasp succeeds in 110 of 120 trials, corresponding to a 91.7% execution success rate.

IV. CONCLUSION

We present a physics-grounded robotic grasp detection framework under low-light conditions, and a real-world low-light RGB-D dataset RLL-Grasp. Current evaluation suggests that physics-grounded illumination-invariant features, together with their adaptive modulation of RGB-D representations, can improve grasp robustness under real-world low-light conditions and support reliable real-world robotic grasping.

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